MIL-A-22681C(SHIPS)
24 March 1972
SUPERSEDING
MIL-A-22681B(SHIPS)
15 September 1970
(See 6.5)

MILITARY SPECIFICATION

ARRESTER, FLASH, MAIN STORAGE BATTERY,

(SUBMARINE USE), GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers flash arresters of the sintered, aluminum oxide, nonwetting type mounted on a hard rubber screw thread base for use with submarine main storage battery cells to prevent flashback of hydrogen explosions or fires into the insides of the cells and to prevent escape of acid spray from within the cells during charge.

1.2 Classification. Flash arresters shall be of the types specified on the applicable specification sheets (see 3.2 and 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-S-801 - Sulfuric Acid, Electrolyte, for Storage Batteries.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carbon Sealing).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-L-10547 - Liners, Case and Sheet Overwraped, Water-Vaporproof or Water-Proof, Flexible.
MIL-C-15816 - Cover, Complete, and Parts, Hard Rubber For Use With Submarine Storage Batteries.
MIL-P-15817 - Parts, Synthetic Rubber, for Use With Batteries or in Battery Compartments of Submarines.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

See Supplement - 1 for list of applicable specification sheets.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6140

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## 3. REQUIREMENTS

3.1 Qualification. Flash arresters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at time set for the opening of bids (see 4.2 and 6.3).

3.2 Detailed requirements for individual types. Detailed requirements or exceptions applicable to individual types shall be as specified in the applicable specification sheet. In the event of any conflict between requirements of this specification and the applicable specification sheets, the latter shall govern (see 6.1).

3.3 Material. Material shall be as specified hereinafter. Material not definitely specified shall be suitable in all respects for the purpose intended.

3.3.1 Dome. Dome material shall be Cortland Grinding Wheel Company, nonwetting, fused, aluminum oxide number A-46-V or Simonds Abrasive Company, blue bond type fused, nonwetting, aluminum oxide, grit size 46, or equal.

3.3.2 Hard rubber parts. Hard rubber parts shall be in accordance with MIL-C-15816.

3.3.3 Soft rubber parts. Soft rubber parts shall be in accordance with MIL-P-15817.

3.4 Shock.

3.4.1 Dome. The dome shall not crack or shatter when subjected to the test of 4.5.1.1 at room temperature.

3.4.2 Complete assembly. The dome, base and sealing of completely assembled flash arresters shall not fail or open in any way when subjected to the test of 4.5.1.2.

3.5 Torque. The seal between the dome and the base shall not fail or open in any way throughout the test of 4.5.2.

3.6 Life. The materials used and the preparation of the seal shall be such that the following performance can be maintained over the required ampere-hours of usage as specified in the applicable specification sheet:

(a) The seal between the dome and the base shall have sufficient adhesion and bond strength that it will not yield and will not permit gas from inside a cell to bypass the dome.
(b) The sealant used will not deteriorate nor in any way enhance corrosion of the dome by the acid electrolyte (see 4.5.3).

3.7 Pressure drop.

3.7.1 Dry flash arresters. Dry flash arresters, at room temperature, shall not cause a pressure drop greater than 5 millimeters of water when subjected to the test specified in 4.5.4.1.

3.7.2 Wet flash arresters. Wet flash arresters, at room temperature, shall not cause a pressure drop greater than 15 millimeters of water when subjected to the test specified in 4.5.4.2. Electrolyte used in this test shall be sulphuric acid in accordance with O-S-801.

3.8 Flash arresting ability. Flash arresters shall prevent ignition of gases on the under (concave) side of the dome during the test specified in 4.5.5.

3.9 Impurities. At the completion of refluxing required in 4.5.6, impurities in the one liter reflux solution shall not exceed the amounts shown in table I.

Table I - Maximum impurities permitted in refluxed solution.

| Material | Weight percent |
|---|---|
| Aluminum | 0.001 |
| Arsenic | None |
| Antimony | None |
| Chlorine | None |
| Nickel | 0.0007 |
| Copper | .037 |



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Table I - Maximum impurities permitted in refluxed solution (cont'd).

| Material | Weight percent |
|---|---|
| Iron | .089 |
| Manganese | .0004 |
| Nitrates (as $NO_3$) | .0148 |
| Platinum | None |
| Selenium | None |
| Sulfites (as $SO_2$) | None |
| Zinc | None |
| Ammonium ($NH_4$) | (Report) |
| Calcium and magnesium oxides | 0.022 |
| Organic and volatile material | .029 |
| Suspended matter | Lead compounds only |

3.10 Silicone resin cure. Silcone resin cure shall be determined by the hot acid test specified in 4.5.7.

3.11 Dimensions. All dimensions for the flash arrester and component parts shall be as specified on the applicable specification sheet.

3.12 Weight. Flash arresters complete with component parts shall not exceed the weight specified on the applicable specification sheet.

3.13 Marking. Flash arresters shall be marked in accordance with MIL-STD-130 and include the manufacturer's name and the month and year of manufacture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.1 and 6.2).

4.2 Qualification tests.[1/] Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center (NAVSEC). Qualification tests shall consist of the examination and tests specified in 4.4 and 4.5.1 through 4.5.7.

4.2.1 Qualification samples. Three sample flash arresters of any type for which qualification is desired shall be submitted. Qualification approval of the type submitted will include qualification of all types. Varying the seal material or method of preparing surfaces for sealing shall be considered to result in a new construction of flash arrester.

4.2.2 Qualification sample data. Each manufacturer submitting qualification samples shall also submit a complete description of his product, which shall include the following:

(a) Names of manufacturers of component parts.
(b) Complete description of sealing compounds, primers and catalysts used to make the seal between dome and base.
(c) Certification that the manufacturer, if his product qualifies, will supply on procurements, flash arresters manufactured with the same materials and by the same methods as those used for the qualification samples.

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).



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4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of 600 flash arresters.

4.3.2 Sampling for examination. Samples for examination (see 4.4), shall be selected in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be 2.5 percent.

4.3.3 Sampling for tests. Four samples from each lot shall be selected at random and subjected to the tests of 4.5.4 through 4.5.7. The complete impurities test of 4.5.6 shall be made on 3 additional samples selected from the first lot of each contract or order. Samples from subsequent lots which are subjected to the impurities test of 4.5.6 shall be analyzed for aluminum only.

4.4 Examination. The samples selected in accordance with 4.3.2 shall be examined to verify conformance with all requirements of this specification which do not involve tests.

4.4.1 Material certification. Certification shall be provided as to conformance with requirements specified in 3.3 through 3.3.3. This certification shall document actual inspection or other verifiable quality data. Copies of the certification documentation shall be forwarded to the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782 and to the procuring activity.

4.5 Test procedures.

4.5.1 Shock tests.

4.5.1.1 Dome. A steel ball weighing 1/2 pound shall be dropped, free fall, from a height of 3 feet onto the center of the top of the dome.

4.5.1.2 Complete assembly. A completely assembled flash arrester shall be subjected to the grade A, class I, type E shock test specified for lightweight equipment in accordance with MIL-S-901. For this test, the flash arrester shall be installed in a submarine cell jar cover which in turn shall be mounted on the anvil plate shown on figure 4 (entitled "Anvil plate of shock-testing machine for lightweight equipment") of MIL-S-901 through the terminal post holes in the cover. Other means of mounting the flash arrester to the shock test machine which duplicate this method may be employed.

4.5.2 Torque test. Twice a week throughout the life test of 4.5.3, the flash arrester shall be completely unscrewed from the cell cover in which it is installed, then it shall be reinstalled in the cover and tightened hand-tight.

4.5.3 Life test. Flash arresters on life test shall be installed on submarine main storage battery cells undergoing test at a laboratory satisfactory to NAVSEC. They shall remain installed on as many groups of cells as may be necessary to cause the flash arresters to be in service for a total of ampere-hours of cell charging and at currents not less than as specified on the applicable specification sheet. They shall be removed every six months, rinsed thoroughly with fresh water, dried for four hours in an oven at a temperature of 125° ± 5°F. and replaced immediately on their respective cells. Upon completion of the life test, flash arresters shall be subjected to the flash arrestment test (see 4.5.5).

4.5.4 Pressure drop test.

4.5.4.1 Dry test. Air shall be vented through a dry flash arrester at the rate specified on the applicable specification sheet. The pressure drop experienced by the air passing through the dome shall be measured with a water manometer.

4.5.4.2 Wet test. Submerge each sample flash arrester in sulphuric acid of 1.265 specific gravity for 2 minutes at an acid temperature of 80°F. No sooner than 15 seconds nor later than 60 seconds after being removed from the acid bath, during this time period the flash arrester shall be in the inverted position to permit drainage, air shall be vented through the flash arrester at the rate specified on the applicable specification sheet. The pressure drop experienced by the air in passing through the dome shall be measured with a water manometer.

4.5.5 Flash arrestment test. While a gas mixture consisting of 67 percent hydrogen and 33 percent oxygen is passed through the flash arrester from the under (concave) side at the rate specified on the applicable specification sheet, an electric spark shall be generated continuously within 1/2 inch of the outside (convex side) of the flash arrester for 15 seconds.



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4.5.6 Impurities test. A flash arrester dome, without hard rubber base, shall be reflexed for 5 hours in one liter of sulphuric acid of 1.265 specific gravity at an acid temperature of 150°F. At the end of the refluxing period, the dome shall be rinsed with distilled water until the rinse water exhibits a pH of 6.5 or higher. The rinse water shall all be collected and added to the refluxed acid. This solution shall then be reduced to a volume of one liter by evaporation and analyzed to determine the amounts of the materials listed in table I (see 3.9).

4.5.7 Hot acid test. A 50-gram sample of flash arrester dome material shall be placed in a 250 ml. beaker filled with 150 ml. of 1.265 $H_2SO_4$. The beaker should be covered and heated at 220°F. for 2 hours. The electrolyte level should be marked at the beginning of the test and maintained throughout by the addition of distilled water. Upon cooling, examine solution for presence of any silica (waxy, gelatinuous, white precipitate) which floats. Presence of any floating silica material shall be cause for rejection.

4.6 Where contracts or orders involve 50 or less flash arresters, quality conformance tests specified in 4.5.4 through 4.5.7 may be waived, provided a certification documenting actual inspection or other verifiable quality data is furnished. In no case shall examination specified in 4.4 be waived.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation and packaging (see 6.1).

5.1.1 Level A. Flash arresters shall be protected and packaged in accordance with MIL-P-116, method III and as specified in 5.1.1.1 and 5.1.1.1.2.

5.1.1.1 Quantities less than 42. Each arrester shall be packaged in a box conforming to one of the following specifications at the option of the supplier:

| Specification | Type or class |
|---|---|
| PPP-B-566 | |
| PPP-B-636 | Domestic |
| PPP-B-665 | |
| PPP-B-676 | |

Box closure shall be as specified in the applicable box specification or appendix thereto. Arresters shall be cushioned to prevent damage during handling, shipment, and storage. Alternatively, flash arresters individually cushioned may be packed in containers for the level of packing specified.

5.1.1.1.2 Quantities of 42 or more. A quantity of 42 flash arresters each individually cushioned and the entire quantity adequately blocked or braced as required to prevent damage from handling, shipment, and storage shall be packed in containers for the level of packing specified.

5.1.1.2 Level C. Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration or physical damage. This level may conform to the supplier's commercial practice.

5.1.2 Packing (see 6.1).

5.1.2.1 Level A. Flash arresters, packaged as specified, shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

| Specification | Type or class |
|---|---|
| PPP-B-585 | Class 3 |
| PPP-B-591 | Weather resistant |
| PPP-B-601 | Overseas type |
| PPP-B-621 | Class 2 |
| PPP-B-636 | Weather resistant |
| PPP-B-640 | Class 2 |



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Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for fiberboard boxes conforming to PPP-B-636, weather resistant class, may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 2 inch wide tape conforming to PPP-T-76. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto.

5.1.2.2 Level B. Flash arresters, packaged as specified, shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

| Specification | Type or class |
|---|---|
| PPP-B-585 | Class 1 or 2 |
| PPP-B-591 | Domestic type |
| PPP-B-601 | Domestic type |
| PPP-B-621 | Class 1 |
| PPP-B-636 | Domestic |
| PPP-B-640 | Class 1 |

Box closures shall be as specified in the applicable box specification or appendix thereto.

5.1.3 Marking. In addition to any special marking, requirements, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

5.2 Use of polystyrene (loose-fill) material.

5.2.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.2.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification and applicable specification sheet.
(b) Type required (see 1.2 and 3.2).
(c) Quality assurance requirements (see 4.1.1).
(d) Preservation, packaging, packing or marking requirements (see 5.1.1).
(e) When polystyrene (loose-fill) material is approved (see 5.2.1).

6.2 Management control system document. The following management control system document should be included on DD Form 1660:

(a) MIL-I-45208 (see 4.1.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-22681 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).



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6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

# 6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 THE MARGINS OF THIS SPECIFICATION HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 6140-N329)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 119-R004

## INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD, FORM 1426 APR 63 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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Naval Ship Engineering Center
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Hyattsville, Maryland 20782

OFFICIAL BUSINESS

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